

Element A

- Identify the need. Move past describing the situation to describe how it is a problem.

Many students with disabilities face difficulty participating in physical and social activities with their peers. These students are often prevented from doing regular activities due to their abilities compared to other students which makes them feel excluded and left out. To promote a healthy school environment, we want to foster inclusivity and engagement for all students of any ability. The need is to develop a portable prototype that will allow students to engage in activities and facilitate more social interaction with other students.

- Define the problem by incorporating relevant research to get a “3” rating.

In our current society and school system, there are many students that face difficulties that prevent them from participating in social and physical activities. As a result, actions need to be taken to help them interact with other people whether it be in the classroom or in public spaces. At least 1 in every 59 children has one or several learning disabilities and that is important because in the US itself, there are more than 5 million children who face difficulties in learning. Specifically the National Center for Educational Statistics shows that 7.3 million students are enrolled in special education within the US itself.

- Who is involved? What are they doing? Where is this happening? When does it occur? Why does it happen? How often does it happen? What unique situations does it occur within, if any? What is the size of the issue?

This solution addresses the problem of inclusion within a school environment. Our solution will help disabled students have a better and easier time interacting with their peers. As a result, the people involved in this would be the disabled students, the educators and assistants for the disabled kids, and the rest of the peers that will interact with our solution and the disabled kid. What the disabled students are doing is interacting with their surrounding environment through accommodations provided by our solution. The place this is happening is at schools throughout Fairfax County, more specifically, inside the gyms at each school. This occurs when P.E. class is active and the students need to get their exercise in. This problem happens because the students with disabilities have an extremely difficult time participating in regular activities due to immobility and other factors. This problem occurs often during every school day and there are not really any unique situations in which this problem would occur since it is quite common. Finally, the size of the issue is quite large since this problem occurs nationwide and worldwide but it also happens across Fairfax County in our local area.

- Use your research to describe your problem in detail. In this element, focus on depth, not breadth. Depth means digging down into the problem, pulling out details that aren't noticeable at first.

According to Pew Research Center and the 2021 US Census Bureau , almost 13% of the US population (42.5 million) have disabilities or special needs. More specifically, 7.3 million students have learning disabilities and are excluded from regular school activities due to their capabilities. The main problem is that students with learning

disabilities often fall behind their peers and involvement in more school activities will improve their overall cognitive abilities and mental health. These disabilities affect students' ability to understand written language and their direct attention. In addition, these students deal with low self-esteem and isolation in the school environment and community.

- **Make sure that the problem is explained in enough detail that anyone could understand the goal of this engineering design problem. Assume your reader knows nothing about the problem or its possible solutions. Include details that you might consider common knowledge.**

The problem is to help disabled students engage in physical education activities with their fellow peers. The disabled students have a difficult time moving with some students being only able to move a single finger. However, the state of Virginia requires every student to participate in physical education class until 11th grade. As a result, it is extremely difficult for both disabled students and their educators to ensure everyone engages in activities. The point of this solution is to extend the field of motion for these students through using assistive and innovative technologies and equipment.

- **How does this problem relate to other contexts? Can solving this problem impact other stakeholders? How can this problem have consequences in other situations.**

This problem relates to other contexts in terms of ethical, medical, and social aspects. Disabilities can hinder development in each of these categories. Disabilities may require solutions that some people regard as unethical while at the same time, disabilities can affect people at a social level, making it harder for them to communicate and socialize with others. Additionally, disabilities can be challenging from a medical standpoint since even with immense advancements in technology, there are still challenges that disabled students face that makes it hard for them to live a "normal" life. While this scenario focuses on one or a couple of schools, the issue of students with learning disabilities is present throughout the whole country and even globally. The solution can impact many other stakeholders such as teachers looking to engage students in physical activities and to foster inclusivity.